How-To Library

Informatica

Performance Tuning Guidelines for MongoDB Connector in Informatica Cloud Data Integration

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Abstract

When you use MongoDB Connector to read data from or write data to MongoDB, you can optimize the performance by tuning the mapping parameters appropriately. This article describes general reference guidelines to help you tune the performance of MongoDB Connector in Cloud Data Integration.

Supported Versions

- Informatica® Cloud Data Integration MongoDB Connector

Table of Contents

Overview. 2
Performance Tuning Parameters. 2
Tune the Heap Size for Secure Agent. 2
Tune the Driver Parameters. 3
Tune the Buffer Size. 3
Tune the Partitioning. 4

Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of MongoDB Connector might vary based on individual environments and other parameters even when you use the same data.

Performance Tuning Parameters

You can optimize the performance of a read or write operation when you use MongoDB Connector by tuning the following parameters:

- Heap Size
- Driver
- Buffer Size
- Partitioning

Tune the Heap Size for Secure Agent

A heap size of 1024MB or more is recommended when you read data from or write data to MongoDB. To improve performance and avoid run-time environment memory issues, configure the Java heap size for Informatica Intelligent Cloud Services Secure Agent.

1. Log in to Informatica Intelligent Cloud Services.
2. Select **Administrator** > **Runtime Environments**.
3. On the **Runtime Environments** page, select the Secure Agent for which you want to increase memory from the list of available Secure Agents.

4. On the upper-right corner of the page, click **Edit**.
5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Server.
6. Edit **JVMOption1** as **-Xmx1024m**. Increase the value based on the packet size you specified.

   The following image shows the **Details** page:



7. Restart the Secure Agent.

Tune the Driver Parameters

You can configure the driver parameters to tune the performance of MongoDB Connector.

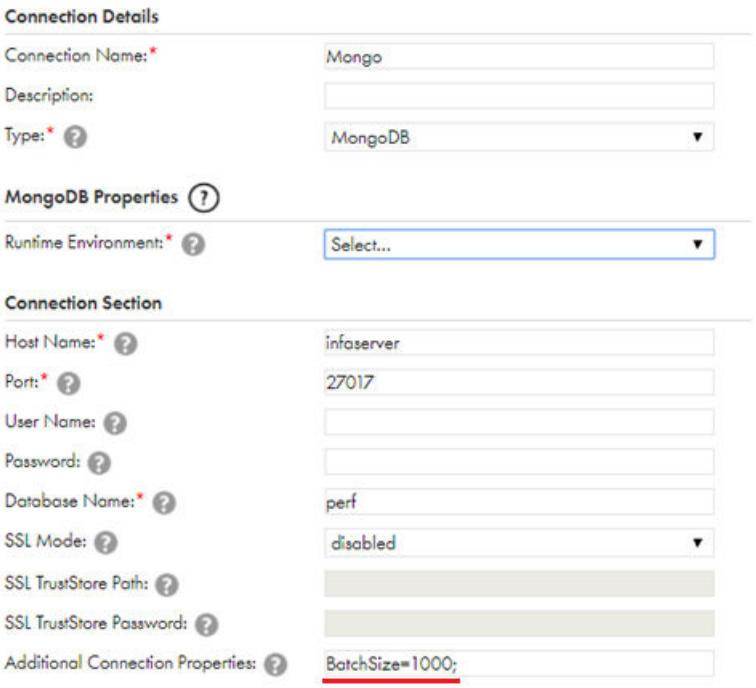
Read Operation

When you configure a MongoDB connection to connect to a MongoDB source, you can specify the `BatchSize` property in the additional connection properties.

Write Operation

When you configure a MongoDB connection to connect to a MongoDB target, you can specify the `DmlBatchSize` property in the additional connection properties.

The following image shows the `BatchSize` property in the MongoDB connection:



For more information, see "MongoDB Connection Properties" in MongoDB Connectors guide under Cloud Data Integration.

Tune the Buffer Size

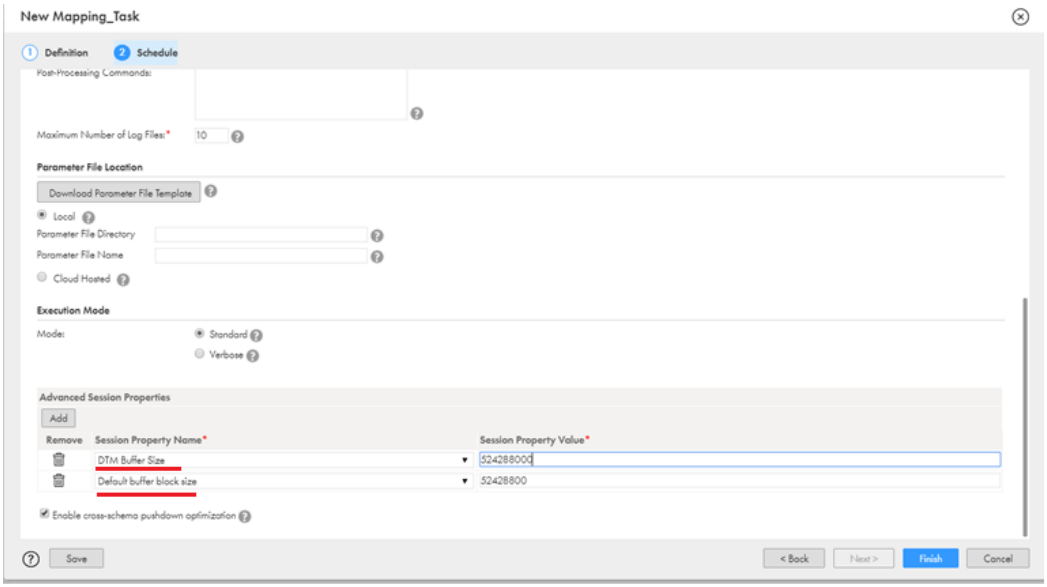
Informatica recommends that you tune the buffer size to improve the performance when you write data to a MongoDB target.

Perform the following steps to tune the buffer size:

1. Create a new mapping task.
2. In the **Schedule** tab, go to **Advanced Session Properties**.
3. Click **Add**.
4. Under **Session Property Name**, select **DTM Buffer Size** and specify a value.
5. Click **Add**.
6. Under **Session Property Name**, select **Default Buffer Block Size** and specify a value.

   Note: The DTM buffer size must be at least 10 times the default buffer block size. The values are in bytes.

   The following image shows the **DTM Buffer Size** and **Default Buffer Block Size** properties:



7. Click **Finish**.

Tune the Partitioning

Read Operation

When you read data from a MongoDB source, you can configure key range partitioning to improve the mapping performance. Specify a minimum heap size of 1024MB and increase if needed when you use partitioning and read data from a MongoDB source.

Write Operation

Partitioning might not improve performance when you write data to a MongoDB target due to a third-party limitation.

Author

Informatica Cloud R&D Team